

SNEHIT VADDI

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SUMMARY

AI Engineer with 5+ years across data engineering, ML systems, and deep learning research, now building production AI agents, eval pipelines, and LLM fine-tuning with enterprise guardrails. Shipped GenAI serving 15,000+ users daily, from multi-agent orchestration to fine-tuned SLMs in HIPAA-regulated environments. Published researcher (SPIE, IEEE) and entrepreneur shipping AI products with 1,000+ active users. Relentlessly curious, driven to ship AI that works for real people.

PUBLICATIONS

- **Can Small Models Reason About Legal Documents? A Comparative Study**, Under Review, [[arXiv, 2026](#)]
- **Do Hallucination Neurons Generalize? Evidence from Cross-Domain Transfer in LLMs**, Under Review, 2026.
- **Detecting E. coli Contamination on Plant Leaf Surfaces Using UV-C Fluorescence Imaging and Deep Learning** [[SPIE, 2025](#)]
- **An Effective Model for Pothole Classification and Admin Alerting System**, [[IEEE ICCMST, 2023](#)]

PROFESSIONAL EXPERIENCE

Oscar Health (Contract)

NYC

Forward Deployed AI Engineer

June 2026 – Present

- **AI Code Review Harness:** Built an org-wide code review harness exposed as an MCP server that flags server-side violations, missing auth scopes, and prompt-injection surface across **900+ pull requests**, with Langfuse tracing on every run. Cut developer review time **30%** and caught **40+ compliance issues** on day 1 before human review.
- **Agentic Analytics Layer:** Built a Cube semantic layer over dbt, exposed as MCP with OAuth2 scope, giving data science teams natural-language access to governed claims metrics. An LLM-as-judge scores answers against a **500-query golden set**, catching metric drift before it reaches users.
- **Agentic Contract Intelligence:** Built multimodal extraction over **14,000+ provider contracts** for the legal team, with agentic self-correction loops that re-query source pages on validation failure or contradictions, lifting first-pass accuracy from **82% to 95%**. A labeled eval harness gates every model and prompt change.
- **AI Enablement & Observability:** Shipped plugins, MCP servers, and skills across legal, finance, and people teams, and rolled Codex out to engineering with admin policies and per-user usage tracking company-wide. Instrumented Prometheus metrics and Langfuse traces across every agent surface with PagerDuty escalation, cutting mean time to detection on degraded tool calls to **under 2 minutes**.

ModMed

Boca Raton, FL (Remote)

AI Engineer - GenAI Applications & LLM Systems

Feb 2025 – April 2026

- **Clinical Ambient AI Scribe:** Shipped clinical AI Scribe serving **15,000+ providers** across 11 specialties (**400K+ daily encounters**), automating 70% of documentation. Built pipeline covering real-time transcription, LLM-powered SOAP generation, and specialty-specific templates.
- **Agentic Document Processing:** Built intelligent document pipeline using OpenAI Agents SDK, OCR, and fine-tuned small vision-language model (Qwen2-VL via HuggingFace) to classify, extract, and route **10M+ clinical pages/month**. Fine-tuned the VLM on domain-specific layouts, achieving superior accuracy over general-purpose models. Replaced \$400K/month vendor with \$20K/month in-house system (**95% cost reduction**).
- **Text2SQL & Semantic Intelligence Layer:** Built clinical knowledge graph over ModMed's EHR warehouse (200+ tables, 11 specialties) with pgvector embeddings, enabling natural language queries over clinical metrics. Reduced analyst request volume by **60%**.
- **Multi-Agent RAG Architecture:** Architected production RAG with LoRA-finetuned SLMs, hybrid pgvector + BM25 retrieval, document parsing, chunking, vectorization, and cross-encoder reranking, achieving **94% retrieval precision** across 50K+ clinical documents.
- **MEDHALT, Hallucination Detection (Open-Source):** Open-sourced clinical hallucination detection suite (DeBERTa NER + LLM-as-judge validation) achieving **92% accuracy** vs. GPT-4. Built evaluation pipelines with MLflow tracking, golden-set regression testing, and automated quality monitoring.
- **Monitoring & Guardrails:** Developed LangChain + LangGraph + Claude monitoring framework for Scribe quality, cutting incident response to under **5 minutes**. Implemented PHI-safe pipelines with PII/PHI redaction, prompt injection filtering,

and audit logging.

- **Databricks & Infrastructure:** Data pipelines on Databricks processing **2M+ records/day** via PySpark and Delta Lake. MLflow for experiment tracking, model registry, and serving. Deployed on Kubernetes (**99.9% uptime**) with CI/CD and A/B experimentation.

GeoSpider AI

AI Software Developer Intern

USA (Remote)

May 2024 – Jul 2024

- **Multi-Agent RAG System:** Built LangGraph-based multi-agent RAG system that **autonomously resolved 65%** of customer tickets across a 50K-doc knowledge base, with dynamic routing, LLM-as-judge scoring, and few-shot prompt selection.
- **Intelligent Routing & Evaluation:** Built LLM-as-judge routing layer with dynamic few-shot prompting. Routes to specialized sub-agents and scores quality, improving **helpfulness from 43% to 76%** and relevance by 30%.
- **Hybrid Search & Inference:** Designed FAISS + keyword hybrid search with semantic reranking achieving **92% recall@10**, serving 150+ concurrent users via vLLM with **40% p95 latency reduction**.
- **Agent Infra & Serving:** Implemented Redis-backed agent memory for multi-turn context management. Built FastAPI inference gateway with model fallback logic, request queuing, and structured logging.

University of Florida

Graduate Researcher, AI/ML

Gainesville, FL

Feb 2023 – Dec 2024

- **Computer Vision Research (Published):** Developed hybrid YOLOv8-ViT model improving small-object detection by **15%** with Grad-CAM/EigenCAM explainability. Published in **SPIE 2025** and **IEEE 2023**. Presented at both conferences.
- **Interactive Analytics & Multi-Modal:** Built React dashboard with Grad-CAM visualizations replacing static reports (**adoption: 15% → 85%**). Prototyped CLIP-based multi-modal retrieval between lab images and research reports.
- **ML Pipeline Automation:** Automated model retraining via MLflow + GitHub Actions CI/CD, cutting deployment from **4 hours to 15 minutes**. Built batch inference with configurable PyTorch/TensorFlow backends.

AT&T (via Accenture)

Software Data Engineer

Bangalore, India

Jun 2021 – Dec 2022

- **Intelligent Dispatch Optimization:** Engineered BERT + XGBoost intent classifier (88% F1) predicting technician dispatch necessity. Eliminated 12K unnecessary dispatches/year, saving **\$2M annually**.
- **Proactive Anomaly Detection:** Built Elasticsearch + Word2Vec anomaly detection system for network telemetry, surfacing incidents and recommending diagnosis paths, **cutting diagnosis time by 40%** and Tier-2 escalations by 25%.
- **Data Engineering at Scale:** Optimized PySpark/Delta Lake pipelines processing **1M+ logs/day** (30% latency reduction). Designed Azure Synapse warehouse with dbt. Built NLP ticket categorization pipeline using fine-tuned BERT (91% accuracy, 500K+ monthly interactions).

TECHNICAL SKILLS

Languages & APIs:	Python, TypeScript, SQL, RESTful APIs, FastAPI, React.js, Streamlit, pandas, scikit-learn
GenAI & Agents:	LangChain, LangGraph, DSPy, OpenAI API, Claude/Anthropic API, MCP (Model Context Protocol), Agent Gateways, Tool Calling, HuggingFace Transformers, PyTorch, RAG, Multi-Agent Orchestration, LoRA/QLoRA Fine-Tuning, Prompt Engineering, vLLM, Guardrails
Evals & Observability:	Promptfoo, Langfuse, LLM-as-Judge, Golden-Set Regression Testing, Hallucination Detection, Drift Detection, Tool Traces, MLflow, Prometheus, Datadog, PagerDuty
Data & Semantic Layer:	Cube, dbt, Databricks (Unity Catalog, MLflow, Delta Lake), PySpark, Apache Spark, Snowflake, Airflow, PostgreSQL/pgvector, FAISS, Pinecone, Qdrant, Elasticsearch, BM25, Cross-Encoder Reranking
Cloud & Infra:	AWS (S3, EC2, Bedrock, Lambda, SageMaker), Azure (Synapse, ML), Docker, Kubernetes, GitHub Actions, CI/CD, Terraform, OAuth2/JWT, Blue-Green Deployment, Auto-Scaling

EDUCATION

University of Florida

Master of Science in Computer & Information Science

Gainesville, FL

Jan 2023 – Dec 2024

GITAM University

Bachelor of Technology in Computer Science (GPA: 3.9/4.0)

Visakhapatnam, India

May 2017 – Jun 2021